

Retail Business Intelligence & Customer Decision Analytics

Final analytical report based on the uploaded 9,800-row retail dataset.

Executive summary

The project upgrades the original Python/Pandas/Excel analysis into a reproducible analytics workflow covering data quality, SQL dimensional modeling, business analysis, customer intelligence, statistical testing, market-basket analysis, forecasting and dashboard design.

Metric	Result
Historical Sales	2,261,536.78
Orders	4,922.00
Customers	793.00
Products	1,861.00
Sub-Categories	17.00
Average Order Value	459.48
Average Shipping Days	3.96
Median Shipping Days	4.00

Key findings

- Historical sales total \$2,261,536.78 across 4,922 orders and 793 customers.
- 2018 sales were \$722,052.02, the highest annual total in the dataset.
- Approximately 28.9% of customers generated 60% of historical sales; approximately 49.6% generated 80%.
- Standard Class averaged 5.01 shipping days versus 2.18 for First Class.
- The Standard-vs-First Class shipping comparison was statistically significant (Mann–Whitney $p < 0.001$).
- Regional shipping duration differences were statistically significant (Kruskal–Wallis $p=0.0076$).
- Order value did not show a statistically significant difference across customer segments ($p=0.4790$).
- The selected forecasting model was ETS Additive Damped with 18.10% holdout MAPE.
- November 2018 was flagged as an unusually high monthly-sales observation by the IQR screening rule.

Business recommendations

- Use RFM segments to differentiate customer retention and reactivation actions.
- Monitor customer concentration because a relatively small customer group contributes a large share of historical sales.
- Investigate high-lift sub-category combinations as candidates for cross-sell experiments; association is not causation.
- Track shipping performance by service mode and region and investigate statistically unusual differences.
- Use the forecast as a planning baseline and retrain/re-evaluate when new monthly data arrives.
- Investigate anomaly periods using operational context before making decisions.

Limitations

The source does not contain profit, cost, quantity, discount, inventory, returns or customer acquisition cost. Therefore the project does not claim profit/margin, discount effectiveness, inventory optimization, return-rate analysis, CAC or true CLV.